AMPACITY	MVA RATING	COND TEMP	SHEATH TEMP	TOTAL LOSS	STATUS
460 A	26.33 MVA	90.0 degC	82.6 degC	44.249 W/m	Converged

1. Design Data & Input Parameters

Parameter	Formula / Symbol	Value	Unit
Conductor material		Cu	
Conductor size		240	mm2
Conductor type		stranded	
Conductor diameter Dc		18.4	mm
Conductor screen OD Dc_sc		19.4	mm
Max conductor temp theta_max		90	degC
Insulation material		XLPE	
Insulation screen OD Di		35.8	mm
Ins outer screen OD Di_sc		36.8	mm
Insulation thickness t_ins		7.70	mm
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.004000	
Screen type		Cu_tape	
Screen mean diameter ds		0.9	mm
Screen area As		35.0	mm2
Armour type		none	
Oversheath material		HDPE	
Cable outer diameter De		44.0	mm
Rated voltage U		33.0	kV
U0 line-to-earth		19.053	kV
Frequency		50	Hz
Bonding		solid	
Formation		trefoil	
Depth to cable centre L		1000	mm
Axial spacing s		44.0	mm
Ambient ground temp theta_a		30	degC
Soil thermal resistivity rho_soil		1.20	K.m/W

2. DC Resistance [IEC 60287-1-1 Cl.2.1.1]

Parameter	Formula / Symbol	Value	Unit
IEC 60228 R0 at 20 degC		7.5400e-05	ohm/m
Temperature coefficient a20		3.9300e-03	1/K
R0 at theta_amb = 30 degC		7.8363e-05	ohm/m
R0 at theta_max = 90 degC		9.6143e-05	ohm/m

3. AC Resistance at theta_max [IEC 60287-1-1 Cl.2.1.2, 2.1.4]

Parameter	Formula / Symbol	Value	Unit
ks strand constant		1.0000	
kp proximity constant		0.8000	
xs4 = [ks(8pif/R0)x10^-7]^2		1.708396	
ys skin effect factor		0.008835	
xp4 = [kp(8pif/R0)x10^-7]^2		1.093373	
yp" intermediate proximity		0.005669	
yp proximity factor		0.004298	
R at theta_max AC = R0(1+ys+yp)		9.7405e-05	ohm/m

4. Dielectric Losses [IEC 60287-1-1 Cl.2.2]

Parameter	Formula / Symbol	Value	Unit
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.004000	
U0 line-to-earth		19.0530	kV
Capacitance C = er/(18 ln(Di_sc/Dc_sc)) 10^-9		0.216938	nF/m
omega = 2 pi f		314.1593	rad/s
Wd = omega C U0^2 tan(delta)		9.8963e-02	W/m
TB880 GP7		Applied	

5. Screen / Sheath Setup [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
Screen material		Cu	
Screen resistivity rho_s		1.7241e-08	ohm.m
Screen area As		35.00	mm2
Screen Rs0 at 20 degC		4.9260e-04	ohm/m

6. Armour Losses [IEC 60287-1-1 Cl.2.4]

Parameter	Formula / Symbol	Value	Unit
Armour type		None	
lam2		0.000000	

7. Thermal Resistances [IEC 60287-2-1]

Parameter	Formula / Symbol	Value	Unit
rho_ins		3.50	K.m/W
rho_osh		3.50	K.m/W
T1 = (rho_ins/2pi) ln(Di_sc/Dc_sc)		0.356632	K.m/W
T2 filler/bedding single-core = 0		0.000000	K.m/W
T3_raw = (rho_osh/2pi) ln(1+2t/ds)		0.204838	K.m/W
T3 x 1.6 factor touching trefoil		0.327741	K.m/W
u = 2L/De		45.4545	
T4 = (rho/2pi) ln(u+sqrt(u^2-1))		0.861297	K.m/W
Sum T = T1+T2+T3+T4		1.545670	K.m/W

8. Sheath Losses at Convergence [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
theta_s at convergence		82.6134	degC
Rs at theta_s		6.1381e-04	ohm/m
X sheath reactance		2.8794e-04	ohm/m
lam1' circulating current		1.136591	
lam1'' eddy current		0.000100	
lam1 = lam1' + lam1''		1.136691	
lam2 armour		0.000000	

9. Ampacity Calculation [IEC 60287-1-1 Cl.1.4]

Parameter	Formula / Symbol	Value	Unit
Delta_theta = theta_max - theta_amb		60.00	K
Etop numerator		59.864683	
Ebot denominator		2.8221e-04	
I = sqrt(Etop/Ebot) exact		460.5770	A
I rounded TB880 GP2		460	A
MVA = sqrt(3) x U x I		26.326	MVA
Iterations		5	
Convergence		YES	

10. Temperature Profile Through Cable

Parameter	Formula / Symbol	Value	Unit
theta_c conductor target		90.0	degC
theta_c back-calculated		90.0000	degC
theta_s sheath surface		82.6134	degC
theta_j cable outer surface		68.1113	degC
theta_a ambient ground		30.0	degC

11. Loss Budget

Parameter	Formula / Symbol	Value	Unit
Conductor Wc = I^2 x R		20.662667	W/m
Sheath Ws = lam1 x Wc		23.487068	W/m
Armour Wa = lam2 x Wc		0.000000	W/m
Dielectric Wd		9.8963e-02	W/m
Total		44.248698	W/m

12. Electrical Parameters

Parameter	Formula / Symbol	Value	Unit
Inductance L		0.916539	uH/m
Reactance X = omega L		287.9394	uOhm/m
Pos-seq resistance R1		208.1247	uOhm/m
Pos-seq impedance Z1		355.2815	uOhm/m
Capacitance C		0.216938	nF/m
Charging current Ic		1.2985	mA/m

Standards: IEC 60287-1-1:2023 | IEC 60287-2-1:2015 | CIGRE TB 880 guidance applied.
GP2: ampacity rounded down. GP6: eddy losses always calculated. GP7: dielectric losses always calculated. GP8: exact T4 formula. T3: 1.6x factor for touching trefoil.
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